ABSTRACT

In order to develop an energy-saving and management system based on the Internet of Things specifically designed for remote computer rooms (RCRs) owned by telecom carriers. The management system based on IoT technology for RCRs is designed. By using technologies such as embedded systems, sensor technology and Java Web, a system realizes equipment management and real-time data visualization of RCRs. Then, a simple but effective air conditioner energy-saving algorithm, called ET algorithm, is developed. Without causing high investment cost, ET algorithm minimizes the energy consumption of air conditioners which accounts for nearly 40% of total consumption, and thus improves the overall energy efficiency of RCRs. Moreover, the proposed system has been implemented at many Telecom operators' RCRs distributed at different regions in Guangdong Province of China, and stably operating for more than two years. Experimental results show that the proposed system not only effectively manages and monitors RCRs, but also achieves 16.64% energy-saving effect.

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LIST OF ABBREVIATIONS

ARM	-	Advanced RISC Machine
CMOS	-	Complementary Metal-Oxide-Semiconductor
CNN	-	Convolutional Neural Network
DCS	-	Distributed Control System
DTMFT	-	Dual Tone Multi-Frequency Technology
EEPROM	-	Electrically Erasable Programmable Read-Only Memory
EMSP	-	E-Mobility Service Provider
ETA	-	Energy Temperature Algorithm
HVAC	-	Heating, Ventilation, and Air Conditioning
IDE	-	Integrated Development Environment
ITER	-	International Thermonuclear Experimental Reactor
LCD	-	Liquid Crystal Display
MPC	-	Model Predictive Control
PCB	-	Printed Circuit Board
PID	-	Proportional-Integral-Derivative
PUE	-	Power Usage Effectiveness
PWM	-	Pulse Width Modulation
RCR	-	Remote Control Rooms
RRU	-	Remote Radio Units
SRAM	-	Static Random-Access Memory
TCP	-	Transmission Control Protocol
TSMCUP	-	Taiwan Semiconductor Manufacturing Company Ultra Power
USB	-	Universal Serial Bus

CHAPTER 1

INTRODUCTION

With the advent of 5G era, energy consumption in operator field has been growing rapidly, due to an explosion in network flow of base station. It is reported that the recognized energy-saving potential for Chinese three major operators in 2014 was approximately 6.71 million tons of standard coal, equivalent to 2% of the total electricity consumption in China. Researchers predict that the annual growth rate of energy consumption in operators' remote computer rooms (RCRs) will reach 7.5%-20%. However, since electricity production mainly relies on burning fossil fuel, increasing energy consumption aggravates CO2 and pollutant emissions, which bring great threat to nature. Therefore, the energy saving and carbon emissions reduction of operators' RCRs is a very meaningful issue.

Generally speaking, a computer room (CR) mainly consists of information technology (IT) equipment (including servers and storages) and other service facilities such as air conditioners (ACs), ongoing power supply, and lighting system. In order to ensure the high-efficient operation of IT equipment, ACs usually works 24/7 to maintain the required indoor temperature. Recent studies indicate that the energy consumed by ACs accounts for nearly 40% of the total energy consumption of a CR. Hence, reducing the energy consumption of ACs is very significant for the energy saving of operators' RCRs. Except for energy-saving issue, the management and maintenance of RCRs, especially for that in suburbs, is another challenging problem for operators. In addition to energy consumption concerns, managing and maintaining remote computer rooms (RCRs), particularly those

located in suburban areas, presents unique challenges for operators. Remote locations often lack easy access to skilled technicians and spare parts, which can lead to prolonged downtime in the event of equipment failure. Moreover, environmental factors such as dust, humidity, and temperature fluctuations can pose additional risks to the reliable operation of IT equipment within these facilities. Implementing robust monitoring systems and proactive maintenance schedules are essential strategies for mitigating these challenges and ensuring the uninterrupted performance of critical IT infrastructure in RCRs.

Without remote management system, engineers usually spend tremendous time and effort on the fault diagnosis of equipment in RCRs distributed across different regions, resulting in huge operation and maintenance cost. Existing researches either focus on reducing the energy consumption of ACs, or on developing management system for RCRs. However, few literatures achieve both energy saving and management purpose of RCRs. It summarizes some representative works from two aspects Energy-saving and Management. For energy savings in ACs, researchers have proposed various schemes such as designing novel energy-conserving equipment, utilizing renewable energy, and adopting intelligent control algorithms, etc. It decreased the energy consumption of a packaged unit AC by using an indirect evaporative cooler to pre-cool air. It improved the energy efficiency of a split-type AC by adding an energy-saving device. This work proposed a fuzzy logic-based thermostat setting strategy to preserve the indoor temperature and humidity and to lower the energy consumption of a central AC compressor. It proposed a fuzzy temperature control scheme for multiple unit ACs. By controlling the operational number of compressors and fans, the energy efficiency can be increased.

Existing work developed a cloud-based AI platform that can run different control algorithms and remotely control ACs. Experimental results show that Fuzzy-PID control and model predictive control (MPC) are the two most energy-efficient control methods. However, the investment cost and implementation difficulty of above methods are much higher than that of controlling ACs' working time. Hence, adjusting the working time of ACs without sacrificing the required temperature is a simple and cost-efficient way to lower the energy consumption of ACs.

For the management of remote CRs or data centers (DCs), existing references have developed some remote management systems for CRs or DCs, which have good management performance but low or even no energy-saving ability. This work presented a CNN-based automatic collection and management system for CRs, which only realized the image acquisition and classification of devices. This work designed an intelligent environment monitoring system to manage and maintain power equipment in DCs. However, this system cannot collect and display energy consumption data of DCs. A cloud green energy management system, proposed. However, they need to improve these behaviors by developing energy saving strategies. It presented a data center infrastructure management system to minimize energy consumption by the coordinated control of equipment. Unfortunately, the reduction rate of total power consumption is relatively low. The proposed a smart monitoring model for DCs based on embedded technology, which realized the management of equipment.

Nevertheless, this model does not involve any energy-saving algorithms. To the best of our knowledge, this study provides the following two contributions we propose an IoT-based management

system for Telecom operators' RCRs. By using technologies such as embedded systems, sensor technology and Java Web, this system enables convenient equipment management and real-time data visualization of RCRs. we design a simple but effective air conditioner energy-saving algorithm, called ET algorithm. Without causing high investment cost, ET algorithm minimizes the energy consumption of ACs by controlling their working time, and thus improves the overall energy efficiency of RCRs.

Economically effective maintenance and monitoring of power systems to ensure high quality and reliability of electric power supplied to customers is becoming one of the most significant tasks of today's power industry. As with any preventive maintenance technology, the efforts spent on the status monitoring are justified by the reduction of the fault occurrence and elimination of consequent losses due to disruption of electric power, damage to equipment, and emergency equipment replacement costs. In the past few years, there have been several significant developments on monitoring technologies for distribution power cables. This review describes technical results relevant to mobile sensing of distributed systems, especially for maintenance tasks. Does not match the resolution of equivalent sensors placed in the direct vicinity of system components.

In addition to sensitivity improvement and subsequent system reliability enhancement, the use of robotic platforms for power system maintenance has many other advantages. Replacing human workers for dangerous and highly specialized operations, such as live maintenance of high-voltage transmission lines, has been a long-standing effort in the power community. Other needs of robotics in power systems include operation in hazardous environments, such as radioactive

locations in nuclear plants, access to tight spaces, such as cable viaducts and cooling pipes, and precise positioning of measurement equipment. Therefore, one may expect that the mobile sensing will play an increasingly important role in the monitoring of power systems.

The Arduino's claim to fame is that all of the hardware and software is open source. The Arduino is an open-source microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs) and 6 analog input. It contains everything needed to support the microcontroller, and it can be simply connected to a computer with a Universal Serial Bus (USB) cable to get started. Alternatively, a more advanced implementation Arduino Mega2560 can be chosen with more input/output pins (54 pins compared to 14 pins of Arduino Uno).

The design described here is compatible with other Arduino boards including the Arduino Mega2560. The Arduino Uno can be programmed with the Arduino Integrated Development Environment (IDE) software based on the Processing IDE. The C-based simple program code for the Arduino is referred to as a sketch. Collections of sketches for specific functionalities are referred to as libraries, often simplifying the programming task of the end-user hiding the code inside the libraries.

The Arduino can be programmed within the limitations of the 32 KB memory and the processing power of the microcontroller. Arduino can function autonomously without being connected to a computer, or alternatively programmed to respond mainly to commands sent from the computer via various software interfaces or to the data acquired from the input channels Additionally, the Arduino can be programmed and controlled via third party programming environments such as

Splash. The hardware functionality of Arduino can be extended with external plug-ins referred as shields commercially manufactured or created by the user.

The project is designed to allow easy use of a mobile phone to control appliances in the telecom. Using a mobile phone the development of the control system will be carried out using IoT. This will communicate with another mobile phone, which in turn controls the devices attached to microcontroller modules. When the action has been carried out then a response is sent to the user. These devices should be controlled as well as turn on/off if required. Most of the times it was done manually. Now it is a necessity to control devices more effectively and efficiently at anytime from anywhere. The underlying principle mainly relies up on the ability of IoT to generate corresponding to a number or code in the number pad and to detect the same number or code from its corresponding IoT Communication.

Whereas mechanization provided human operators with machinery to assist them with the muscular requirements of work, automation greatly reduces the need. The technology used here is embedded system with mobile technology which is future of today's modern electronics. Microcontroller is used here for centralized operation and digital processing. A microcontroller can be considered a self-contained system with a processor, memory and peripherals and can be used as an embedded system. Microcontrollers are used in automatically controlled products and devices, such as automobile engine control systems, implantable medical devices, remote controls, office machines, appliances, power tools, and toys.

By reducing the size and cost compared to a design that uses a separate microprocessor, memory, and input/output devices,

microcontrollers make it economical to digitally control even more devices and processes. We have also incorporated a special feature of voice acknowledgment with the help of apr9600. The apr9600 device offers true single-chip voice recording, non-volatile storage, and playback capability for 40 to 60 seconds. Thus, the project has become more users friendly and also efficient. The care has been taken while developing its hardware and software part that it should be compatible with any kind of mobile handset.

1.1 WHY TO USE MOBILE TECHNOLOGY?

Mobile technology is a wireless technology, which is the major reason to make it handy and user friendly. The wide spread mobile technology now-a-days help to work even in remote areas. It is very much economical to get service of this technology, as every service provider is in race of giving best service in least charge of money. Even in the developing country like india, near about 60-70% of population is familiar to mobile technology and actually using it. This proportion is supposed to increase very drastically.

In this particular project, there is no any specification on the type and the owner company of your mobile handset. This increases the flexibility of project. The aim of the proposed system is to develop a cost-effective solution that will provide controlling of home appliances remotely and enable home security against intrusion in the absence of homeowner. The system provides availability due to development of a low cost system.

The home appliances control system with an affordable cost was thought to be built that should be mobile providing remote access to the appliances and allowing home security. Though devices connected as home and office appliances consume electrical power. Most of the

times it was done manually. Now it is a necessity to control devices more effectively and efficiently at anytime from anywhere. In this system, we are going to develop a cellular phone-based home/office appliance. This system is designed for controlling arbitrary devices, it includes a cell phone (not included with the system kit, end user has to connect his/her cell phone to the system) which is connect to the system via head set. To active the cellular phone unit on the system a call is to be made and as the call is answered, in response the user would enter a two/three-digit password to access the system to control devices.

The implementation of password-protected device control via relays offers a secure and convenient method for remotely managing critical systems. By restricting access to specific individuals who possess the designated passwords, operators can maintain tight control over device operation while safeguarding against unauthorized manipulation. Furthermore, integrating telephony-based notification systems adds an extra layer of functionality, enabling real-time alerts to be sent to multiple designated recipients, ensuring swift response to any event or anomaly detected within the system. This seamless combination of password-protected control and telephony-based alerting enhances operational efficiency and security in diverse environments, from industrial facilities to commercial enterprises.

Overall, the integration of specific passwords, relays for device switching, and the alarm unit with remote communication capabilities reinforces the system's security, reliability, and functionality. By combining these features, the system provides a robust framework for remote monitoring, control, and event management, catering to the diverse needs of users across various applications and industries.

CHAPTER 2

LITERATURE REVIEW

[1] "Is technological innovation effective for energy saving and carbon emission reduction" by B. Li, S. Yin and M. Zhang (2020)

TITLE: Is technological innovation effective for energy saving and carbon emissions reduction

YEAR: 2020

AUTHOR: B. Li, S. Yin and M. Zhang

DESCRIPTION:

This project investigates how technological innovation impacts energy conservation and carbon emissions reduction through thorough analysis and empirical research. By examining data on energy consumption and emissions trends, it identifies avenues for innovation to enhance sustainability. The findings will inform policymakers, industry leaders, and environmental advocates, guiding future innovation toward a more sustainable trajectory.

REMARK:

This project commendably addresses environmental challenges by examining the role of technological innovation in energy conservation and carbon emissions reduction. Through meticulous analysis and empirical research, it offers valuable insights into sustainable practices, positioning itself as a significant contribution to the field. However, it's crucial to acknowledge the complexity of the issues and ensure recommendations are practical and inclusive. Overall, this project advances our understanding of how technological innovation can promote a sustainable future.

[2] "Comparison of different solar assisted air conditioning systems for australian office buildings" by L. Guan, Y. Ma, W. Miller, and S. Saha (2016)

TITLE: Comparison of different solar assisted air conditioning systems for australian office buildings

YEAR: 2016

AUTHOR: L. Guan, Y. Ma, W. Miller, and S. Saha

DESCRIPTION:

This project compares various solar-assisted air conditioning systems designed for Australian office buildings, aiming to pinpoint the most efficient solutions for the region's specific climate and energy needs. By evaluating factors like performance, cost-effectiveness, and environmental impact, it offers valuable guidance on selecting and implementing sustainable cooling systems. The outcomes are anticipated to benefit decision-makers, architects, and industry experts, aiding in the widespread adoption of energy-efficient cooling technologies.

REMARK:

This project is pivotal in advancing sustainable building practices, specifically in addressing energy consumption and environmental impact in Australian office buildings. Focusing on solar-assisted air conditioning systems, it targets a key area for reducing carbon emissions and improving energy efficiency. Through a comparative approach, it provides a thorough understanding of system strengths and limitations, aiding informed decision-making. However, ensuring scalability, low maintenance, and user comfort are vital for the long-term success of recommended systems. Overall, this project offers significant promise in mitigating climate.

[3] "Smart monitoring embedded service for energy-efficient and sustainable management in data centers" by V. Gilart-Iglesias, J. Gil-Martínez-Abarca, D. Marcos-Jorquera, and F. Mora-Gimeno,(2016)

TITLE: Smart monitoring embedded service for energy-efficient and sustainable management in data centers

YEAR: 2016

AUTHOR: V. Gilart-Iglesias, J. Gil-Martínez-Abarca, D. Marcos-Jorquera, and F. Mora-Gimeno

DESCRIPTION:

This project introduces a smart monitoring embedded service for energy efficient and sustainable management in data centers. By integrating advanced monitoring technologies, it optimizes energy usage and enhances sustainability practices. Real-time monitoring enables proactive management, identifying inefficiencies to reduce environmental impact and revolutionize data center operations.

REMARK:

This project represents a significant advancement in addressing the environmental impact of data centers, which are notorious energy consumers. Real-time monitoring and proactive management capabilities enable data center operators to identify inefficiencies promptly and take corrective actions, contributing to overall sustainability goals. However, it's essential to consider factors such as scalability, compatibility with existing infrastructure, and data security to ensure successful implementation and adoption. Overall, this project holds promise in revolutionizing data center management practices towards greater energy efficiency and sustainability.

CHAPTER 3
SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The remote angular position control system of a stepper motor using DTMF technology offers a novel approach to communication and control, especially in scenarios where traditional radio frequency (RF) methods may be impractical or costly. By leveraging DTMF technology through mobile phone networks, the system allows for seamless remote control of the stepper motor's angular position from anywhere in the world. This innovation not only provides simplicity and audibility but also enhances accessibility and convenience for operators. One notable advantage of this system is its simplicity and cost-effectiveness. Unlike traditional RF communication systems that require additional transmitting and receiving devices, the use of DTMF technology relies solely on standard mobile phones, making the setup rugged and straightforward.

Additionally, the integration of DTMF technology enhances system reliability by providing more reliable communication channels, reducing the susceptibility to signal interference or environmental noise that may affect traditional signaling methods. This increased reliability translates to improved responsiveness and reduced downtime, crucial factors in applications where timely intervention is paramount.

Furthermore, the remote-control capabilities afforded by DTMF technology offer unprecedented convenience and flexibility for operators, enabling them to execute commands or adjustments remotely without the need for physical presence. This not only enhances operational efficiency but also contributes to cost savings by

minimizing travel expenses and optimizing resource allocation. Moreover, the versatility of DTMF technology extends beyond basic control functions, allowing for more sophisticated command sequences and automation routines to be executed remotely.

Overall, the continuous evolution of this system addresses the dynamic demands of modern communication and control systems effectively. Its integration with IoT-enabled energy management further underscores its versatility and relevance in empowering telecom operators to achieve cost savings and environmental sustainability while optimizing network performance.

3.2 PROPOSED SYSTEM

In the proposed IoT system, sensors are smart meters and temperature modules. And they are used to collect electricity usage and temperature of RCRs, respectively. While an infrared module, as the actuator, is adopted to learn infrared instructions of different air conditioners and to control them, so as to adjust the indoor temperature of RCRs. The prototype and design of each sensor and actuator will be described in detail in the next section. Based on Web and database technology, EMSP is developed, which mainly includes four modules System management module, Intelligent information module, Remote control module, and Alarm module.

The system management module encompasses various crucial components, including robust data encryption protocols to ensure the security and confidentiality of sensitive information stored within the database. Access control policies are implemented to regulate user permissions, restricting access based on predefined roles and privileges. Audit trail mechanisms are integrated to track and record all

system activities, enhancing accountability and facilitating forensic analysis in case of security breaches.

Real-time monitoring capabilities allow administrators to monitor system performance, detect anomalies, and respond promptly to any issues, ensuring optimal functionality and uptime. Furthermore, seamless integration with external systems or devices enhances the system's versatility and interoperability, facilitating broader application in diverse environments and industries. Comprehensive user training and ongoing support services are provided to ensure proficient utilization of the platform by system administrators and operators, maximizing its potential benefits and minimizing operational disruptions.

The system is designed with scalability and flexibility in mind, enabling easy expansion or customization to accommodate evolving requirements and technological advancements. Moreover, adherence to relevant regulatory standards and industry best practices ensures compliance with legal requirements and quality standards, fostering trust and confidence among stakeholders while mitigating potential risks or liabilities.

However, ordinary users can only look over the information of some specific RCRs. Intelligent information module mainly offers map display, real-time and historical data, and device information. The geographical distribution of all RCRs using the proposed system will display in Map. Users can examine device information to ensure each device operates normally. Moreover, real-time and historical data such as power, voltage, current, electricity consumption and indoor temperature are available for analyzing energy efficiency.

According to the real-time operation situation of a RCR, administrators can remotely and manually turn on/off ACs and adjust temperature set point of ACs. Besides, engineers can also analyze historical electricity usage and predict future energy consumption according to the change of climatic conditions and indoor temperature, so as to design an automatic energy-saving control strategy to reduce energy consumption and liberate manpower. Alarm module is responsible for publishing real-time alarm information and checking historical alarm records such as temperature abnormality, PUE overload, and ARM embedded system disconnected, etc.

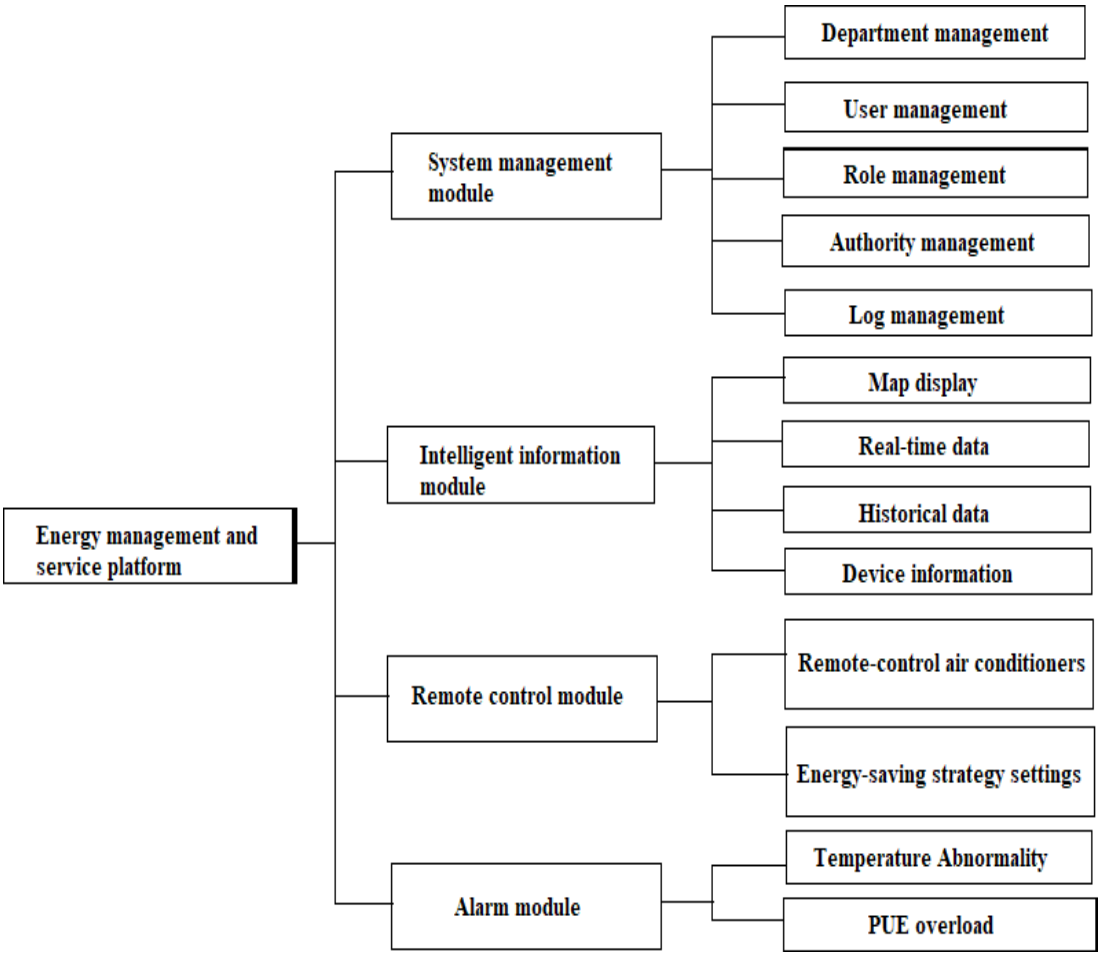


Figure 3.1 Overall Framework of EMSP.

CHAPTER 4

HARDWARE REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

- Arduino Microcontroller
- Voltage regulator
- Gas sensor
- ESP32
- Temperature sensor

4.1.1 Arduino Microcontroller

The Arduino Uno is a versatile microcontroller board built around the ATmega328P chip, offering a robust platform for electronic projects. Featuring 14 digital input/output pins, including 6 PWM outputs, and 6 analog input pins, it provides ample connectivity options. With power supplied via USB or an external source, it's easily programmable using the Arduino Software. The Arduino Uno's user-friendly interface and extensive documentation make it accessible for beginners delving into electronics projects. Expansion capabilities are vast, with numerous shields available for added functionalities like Ethernet connectivity or motor control. Its compatibility with a wide range of sensors and actuators, coupled with extensive community support, makes it an ideal choice for beginners and experienced developers alike. The Arduino Uno's simplicity, versatility, and robust ecosystem have solidified its position as a staple in the DIY electronics and prototyping community.



Figure 4.1 Arduino Microcontroller

4.1.2 Arduino Specialization

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	MHz

Table 4.1 Arduino Specification

4.1.3 Voltage Regulator

A voltage regulator maintains a steady output voltage level despite variations in input voltage or load conditions. ICs regulator is mainly used in the circuit to maintain the exact voltage which is followed by the power supply. A regulator is mainly employed with the capacitor connected in parallel to the input terminal and the output terminal of the IC regulator. A regulator is mainly employed with the capacitor connected in parallel to the input terminal and the output terminal of the IC regulator. Voltage regulators come in various forms, including linear regulators and switching regulators. Voltage regulators can be integrated into integrated circuits (ICs) or used as standalone modules, offering flexibility in design and implementation. As we require a 5V we need LM7805 Voltage Regulator IC.

- Input voltage range 7V- 35V
- Current rating $I_c = 1A$
- Output voltage range $V_{Max=5.2V}, V_{Min=4.8V}$

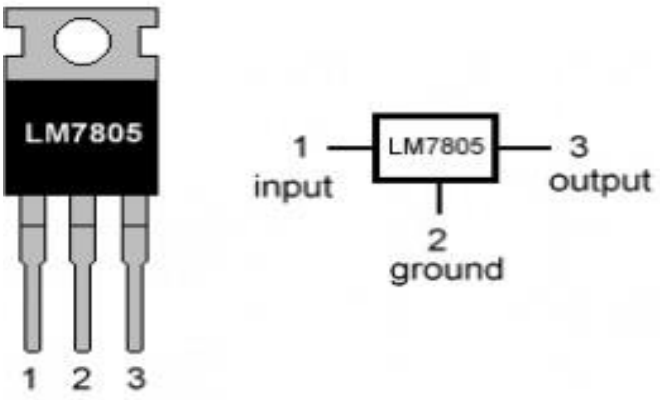


Figure 4.2 Voltage Regulator

4.1.4 Gas Sensor

A gas detector is a device which detects the presence of various gases within an area, usually as part of a safety system. this type of equipment is used to detect a gas leak and interface with a control system so a process can be automatically shut down. a gas detector can also sound an alarm to operators in the area where the leak is occurring, giving them the opportunity to leave the area. This type of device is important because there are many gases that can be harmful to organic life, such as humans or animals. Gas detectors are usually battery operated. they transmit warnings via a series of audible and visible signals such as alarms and flashing lights, when dangerous levels of gas vapors are detected. as detectors measure a gas concentration, the sensor responds to a calibration gas, which serves as the reference point or scale. as a sensor's detection exceeds a preset alarm level, the alarm or signal will be activated. as units, gas detectors are produced as portable or stationary devices. originally, detectors were produced to detect a single gas, but modern units may detect several toxic or combustible gases, or even a combination of both types.



Figure 4.3 Gas Sensor

4.1.5 ESP32

ESP32 is a single 2.4 GHz Wi-Fi-and-Bluetooth combo chip designed with the TSMC ultra-low-power 40 nm technology. It is designed to achieve the best power and RF performance, showing robustness, versatility and reliability in a wide variety of applications and power scenarios. The ESP32 series of chips includes ESP32-D0WD-V3, ESP32-D0WD, ESP32-D0WDQ6, ESP32-D2WD, and ESP32-S0WD, ESP32-D0WD-V3 and ESP32-D0WDQ6-V3 are based on ECO V3 wafer. ESP32 is a highly-integrated solution for Wi-Fi-and-Bluetooth IoT applications, with around 20 external components. ESP32 integrates an antenna switch, RF balun, power amplifier, low-noise receive amplifier, filters, and power management modules. As such, the entire solution occupies minimal Printed Circuit Board area. ESP32 uses CMOS for single-chip fully-integrated radio and baseband, while also integrating advanced calibration circuitries that allow the solution to remove external circuit imperfections or adjust to changes in external conditions. The mass production of ESP32 solutions does not require expensive and specialized Wi-Fi testing equipment.

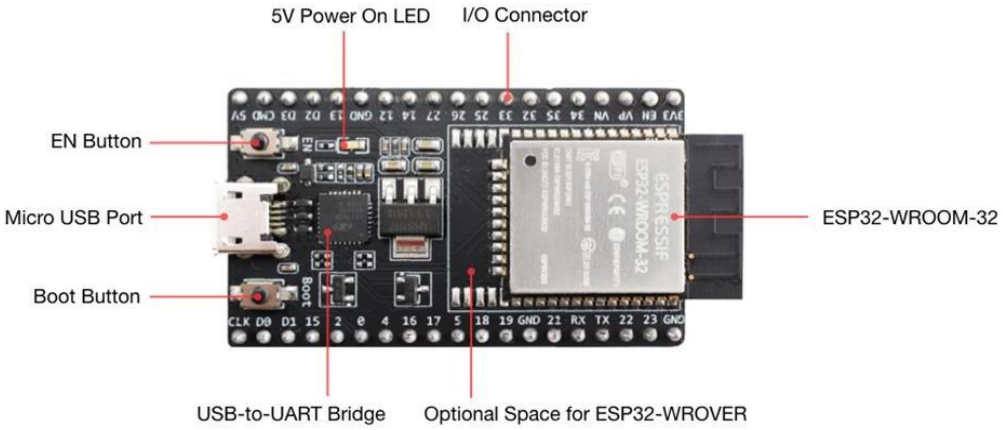


Figure 4.4 ESP32

4.1.6 Temperature Sensor

A temperature sensor is a device that gathers data concerning the temperature from a source and converts it to a form that can be understood either by an observer or another device. These sensors come in many different forms and are used for a wide variety of purposes, from simple home use to extremely accurate and precise scientific use. These devices typically take one of two forms are contact sensors measure their own temperatures after they have achieved thermal equilibrium with their environments, and noncontact sensors measure heat radiation from their environments within a given area.

- Temperature range: -40 degrees C to 150 degrees C / -40 degrees F to 302 degrees F
- Output range: 0.1V (-40 degrees C) to 2.0V (150 degrees C) but accuracy decreases after 125 degrees C
- Power supply: 2.7V to 5.5V only, 0.05 mA current draw.

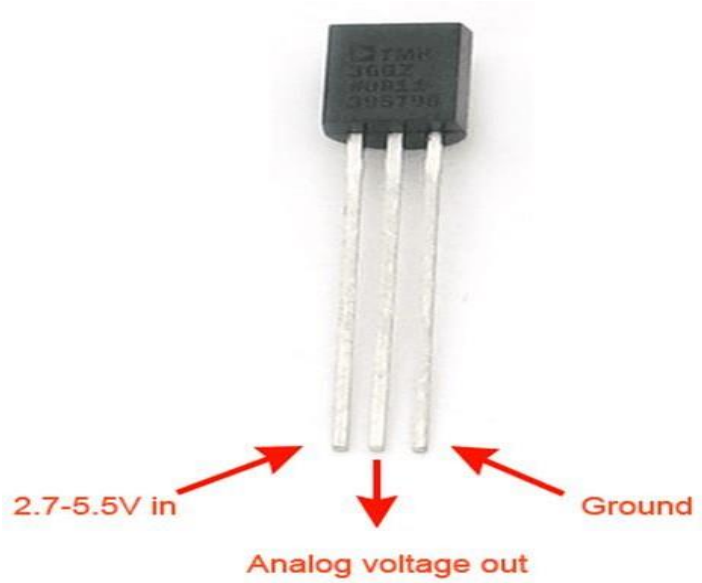


Figure 4.5 Temperature Sensor

CHAPTER 5

SOFTWARE REQUIREMENTS

5.1 SOFTWARE REQUIREMENTS

- Arduino IDE

5.1.1 Arduino IDE

Arduino Integrated Development Environment (IDE) stands as a beacon in the realm of electronics prototyping, offering a comprehensive platform for programming Arduino microcontroller boards. Arduino IDE serves as a gateway for enthusiasts and professionals alike, providing a user-friendly interface coupled with powerful features to unleash creativity in the realm of embedded systems and electronic projects.

Arduino IDE encapsulates a plethora of tools and utilities tailored to streamline the development process. At its core lies a robust text editor, equipped with syntax highlighting and auto-completion capabilities, facilitating seamless code composition. The compiler, an integral component of the IDE, translates human-readable code into machine-readable instructions, bridging the gap between conception and execution. The versatility of Arduino IDE extends beyond its core functionalities, encompassing a myriad of features catering to diverse project requirements. The built-in serial monitor emerges as a crucial tool for debugging and real-time communication with Arduino boards, expediting the troubleshooting process and enhancing project robustness. Additionally, Arduino IDE offers comprehensive support for managing libraries, boards, and preferences, enabling seamless integration of third-party hardware.

The utility of Arduino IDE transcends traditional boundaries, finding application across a spectrum of domains In educational contexts, Arduino IDE facilitates hands-on learning experiences, empowering students to grasp complex concepts through practical experimentation. In industrial applications, Arduino IDE drives the development of bespoke solutions tailored to address specific challenges, ranging from automation and control systems to environmental monitoring and data logging.

The practical implications of Arduino IDE, illustrating its transformative impact on real-world projects. Consider a Smart Home Automation System powered by Arduino boards programmed via Arduino IDE. By leveraging the flexibility and versatility of Arduino IDE, developers can design intuitive interfaces to control lights, appliances, and security systems, ushering in a new era of home automation. Similarly, an Environmental Monitoring Station equipped with Arduino-based sensors and data logging systems programmed using Arduino IDE empowers researchers to gather invaluable insights into air quality, temperature, and humidity levels.

Arduino IDE emerges as a cornerstone in the landscape of electronics prototyping and project development, democratizing access to technology and fostering a culture of innovation. Through its intuitive interface, robust features, and extensive community support, Arduino IDE empowers creators and innovators to transform ideas into reality, driving progress and shaping the future of embedded systems and IoT. As we navigate the ever-evolving landscape of technology, Arduino IDE stands as a beacon of inspiration, illuminating pathways to creativity and discovery.

CHAPTER 6
WORKING PRINCIPLE

6.1 BLOCK DIAGRAM

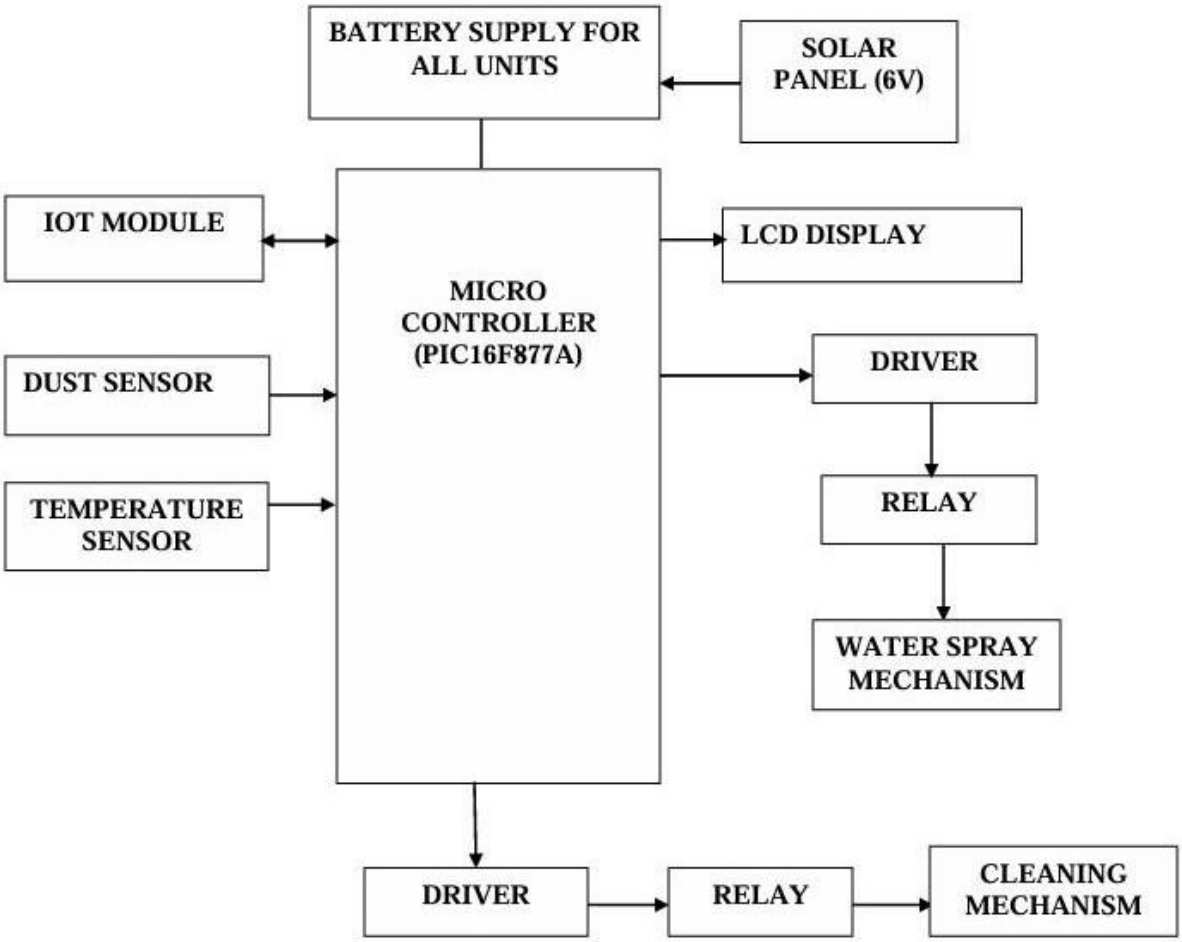


Figure 6.1 Block Diagram

In order to achieve the energy saving and management, this article proposes an IoT-based energy-saving and management system for Telecom operators' remote computer rooms (RCRs). Firstly, a management system based on IoT technology for RCRs is designed. By using technologies such as embedded systems, sensor technology and IoT, this system realizes equipment management and real-time data

visualization of RCRs. Then, a simple but effective air conditioner energy-saving algorithm, called ET algorithm, is developed. Without causing high investment cost, ET algorithm minimizes the energy consumption of air conditioners which accounts for nearly 40% of total consumption, and thus improves the overall energy efficiency of RCRs.

In the present project a microcontroller is used as a control unit which gets inputs (instructions, commands) from a mobile connected through IoT. To make the connection more secure, consumer authentication along with a password will be provided. To switch on/off any appliance positioned at controller's part, the cellular phones are connected; the appropriate tone and password are entered. The tone entered is decoded via the IoT which further translates it into binary values. Binary values are the input to the microcontroller which verifies each one individually and corresponding output is given at the output terminal. Thus, when the relay drive is activated by the microcontroller, the device either gets on or is switched off as per the requirement. Our project makes use of auto answer facility and hence eliminates the need of a ring detector circuit.

The Internet of Things (IoT) has been increasingly utilized in the energy sector for efficient management and conservation. When it comes to telecom operators and remote cell sites (RCR), IoT can play a crucial role in energy saving and management. Then the work using Blynk server as a platform:

Sensor Deployment: Sensors are installed at various points within the telecom infrastructure, including cell towers, base stations, data centers, and other relevant locations. These sensors can monitor parameters such as energy consumption, temperature, humidity, power usage, equipment status, etc.

Data Collection: The sensors collect real-time data on the monitored parameters. This data is then transmitted to a central server or cloud platform. In this case, Blynk server could serve as the platform for data collection due to its compatibility with IoT devices and its ability to handle data securely.

Data Processing and Analysis: The data collected from the sensors is processed and analyzed to gain insights into energy consumption patterns, identify areas of inefficiency, detect anomalies, and predict future energy needs. Advanced analytics algorithms can be employed to perform this analysis efficiently.

Energy Management and Optimization: Based on the insights obtained from the data analysis, energy management strategies are formulated. These strategies may include optimizing the operation of telecom equipment, adjusting power usage based on demand fluctuations, scheduling maintenance tasks to minimize downtime, and implementing energy-saving measures such as turning off non-essential equipment during off-peak hours.

Remote Monitoring and Control: Using the Blynk mobile or web application, telecom operators can remotely monitor the energy consumption and performance of their infrastructure in real-time. They can also remotely control devices and equipment to implement energy-saving measures or address any issues that may arise.

Continuous Improvement: The Blynk mobile and web applications serve as intuitive interfaces, granting operators access to comprehensive dashboards, alerts, and notifications. By implementing IoT-based energy saving and management solutions using the Blynk server, telecom operators can achieve significant cost savings, improve operational efficiency, and reduce their environmental footprint.

CHAPTER 7

RESULTS AND DISCUSSION

The implementation of IoT energy-saving and management solutions for telecom operators and Remote Radio Units (RRUs) utilizing a Blynk server has yielded significant outcomes, underscoring the transformative potential of IoT technology in optimizing energy consumption and enhancing operational efficiency within the telecommunications sector. Through the integration of IoT-enabled sensors and devices, real-time monitoring of energy usage across telecom infrastructure was achieved, providing stakeholders with comprehensive insights into power consumption patterns and environmental conditions. The data served as a foundation for implementing targeted energy optimization strategies, including the optimization of HVAC systems, scheduling of equipment shutdown during off-peak hours, and load balancing to ensure optimal resource utilization. One of the key advantages of the IoT-based solution was the ability to remotely manage and control telecom infrastructure from a centralized platform. Through the user-friendly interface of the Blynk mobile application or web dashboard, operators could access real-time data, receive alerts for abnormal energy consumption or equipment failures, and initiate corrective actions remotely. The scalability of the IoT solution allows for seamless integration with additional sensors, devices, and telecom infrastructure components, enabling continuous optimization and refinement of energy management strategies. Future prospects in the area include advancements in IoT technology, the potential to further enhance the efficiency and effectiveness of energy-saving initiatives in the telecommunications sector.

Energy Consumption Monitoring:

Through the implementation of IoT-enabled sensors and devices, real-time monitoring of energy consumption across telecom infrastructure was achieved. The integration of energy meters and environmental sensors facilitated the collection of data pertaining to power usage, temperature, humidity, and other relevant parameters. The data was transmitted to the Blynk server, providing stakeholders with comprehensive insights into energy utilization patterns and enabling proactive decision-making.

Energy Optimization Strategies:

By leveraging the data collected through IoT sensors, telecom operators and RRU managers were able to identify areas of inefficiency and implement targeted energy optimization strategies. The included the optimization of HVAC systems based on environmental conditions, scheduling of equipment shutdown during off-peak hours, and load balancing to ensure optimal resource utilization. Through the utilization of Blynk's user-friendly interface, stakeholders could remotely monitor and control energy-consuming devices, thereby minimizing wastage and maximizing energy savings.

Remote Management and Control:

One of the key advantages of the IoT-based solution was the ability to remotely manage and control telecom infrastructure from a centralized platform. Using the Blynk mobile application or web dashboard, operators could access real-time data, receive alerts for abnormal energy consumption or equipment failures, and initiate corrective actions remotely. It is not only enhanced operational efficiency but also reduced the need for on-site interventions, thereby lowering maintenance costs and improving overall system reliability.

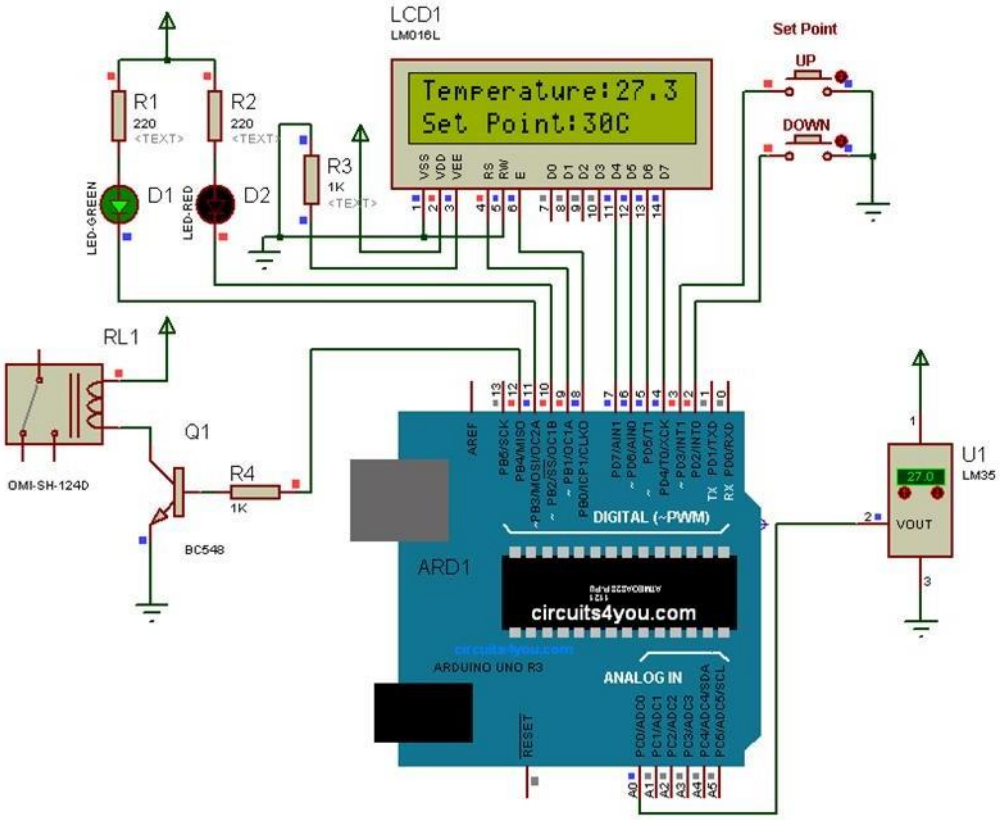


Figure 7.1 Stimulated Diagram

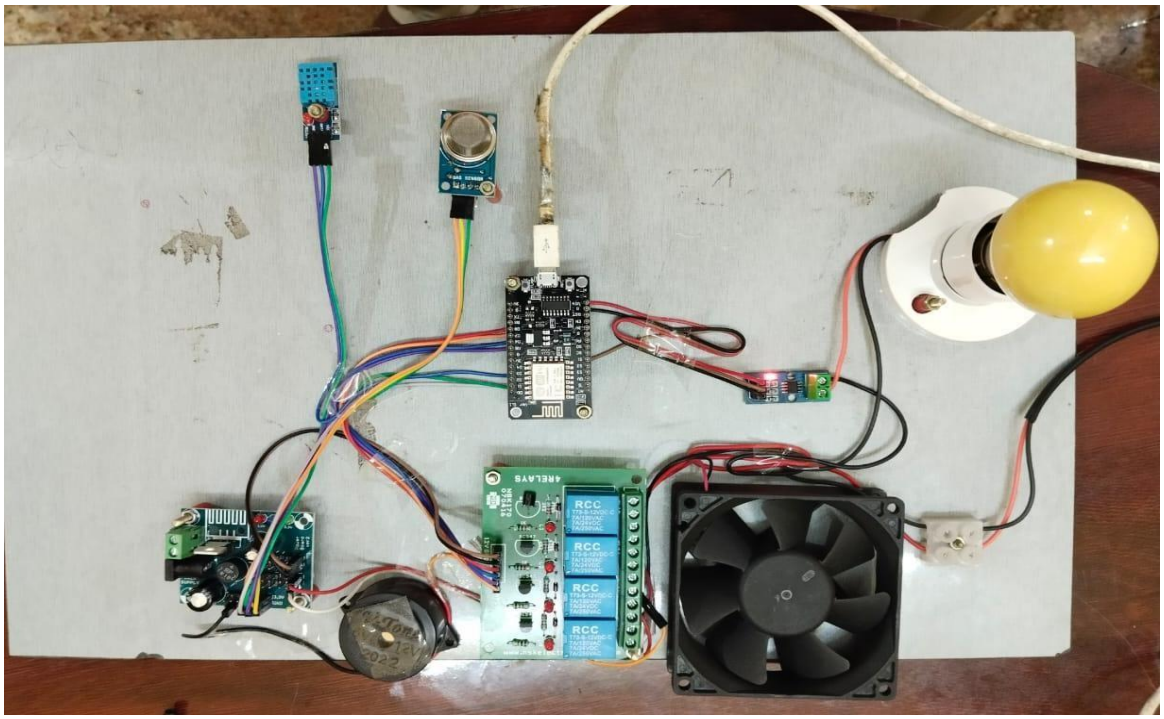


Figure 7.2 Connection Diagram

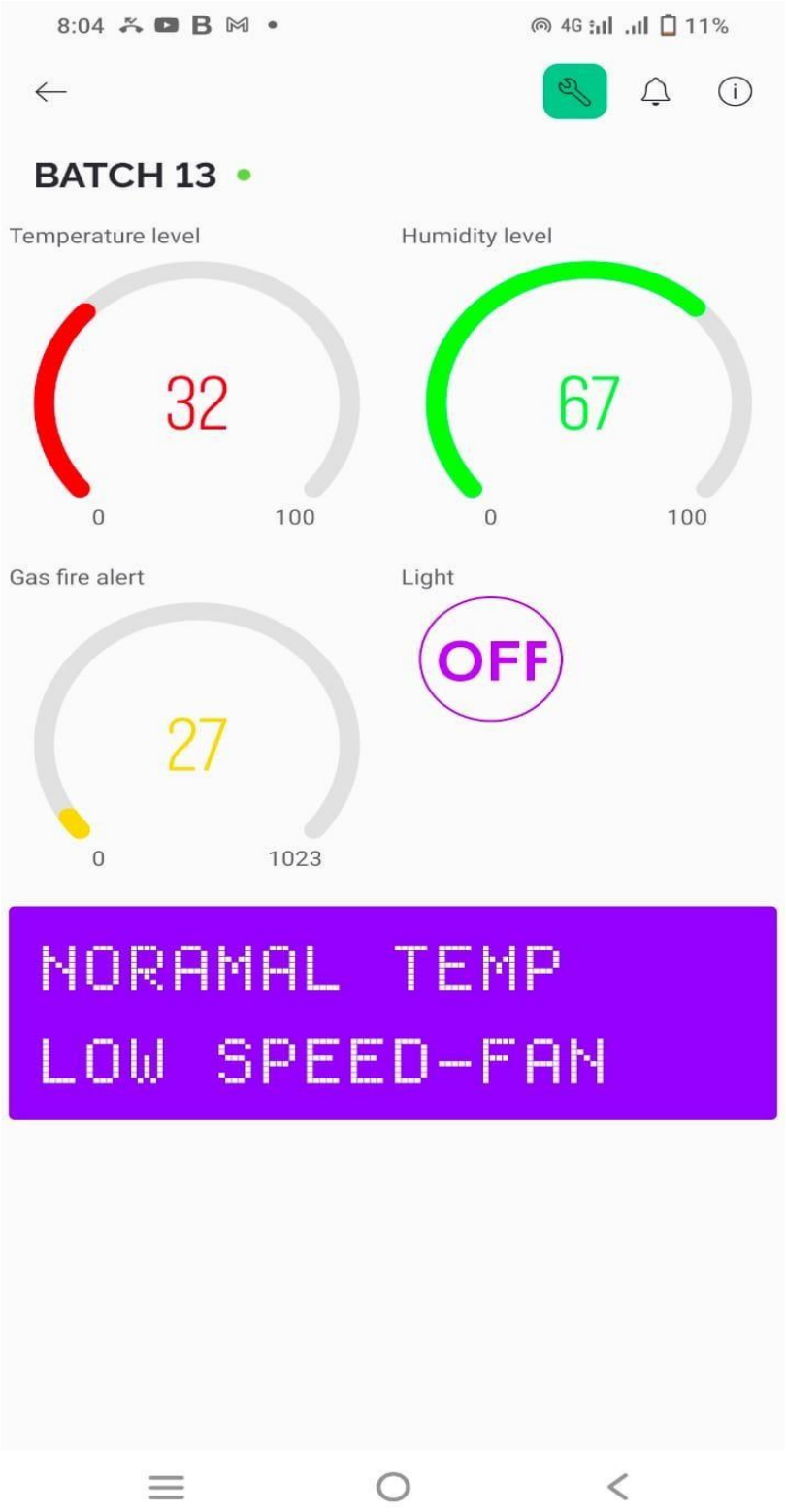


Figure 7.3 Mobile Web Dashboard

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION AND FUTURE ENHANCEMENT

The work presents an IoT system to manage Telecom operators' remote computer rooms and to save their energy. Smart meter, temperature module, infrared module and embedded control system are applied as the hardware platform, while EMSP is regard as the software platform. Furthermore, a simple but effective ET energy-saving algorithm is proposed to control the switching frequency and working time of air conditioners, which maximizes their energy efficiency. Numerous operation results during 2018-2019 demonstrate that the proposed system not only successfully manages the remote computer room but also effectively reduces its total energy consumption, and has promising engineering application value. In the future, we will develop fuzzy control algorithm to further enhance energy-saving rate and apply the proposed system to large data centers or buildings. And wireless technology such as IoT and Wi-Fi will be employed to replace TCP/IP and RS485 communication, thus avoiding wiring inconvenience and providing better remote computer rooms' management. Embracing wireless technologies such as IoT and Wi-Fi for communication purposes will streamline installation and maintenance processes while offering improved remote management capabilities. These advancements promise to enhance the operational efficiency and sustainability of remote computer room management, paving the way for a more resilient and environmentally conscious approach in the telecommunications industry.

APPENDIX

DESIGN CODE:

```
#define BLYNK_TEMPLATE_ID "YOUR_TEMPLATE_ID"
#define BLYNK_TEMPLATE_NAME "Telecom"
#define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>
// You should get Auth Token in the Blynk App.
// Go to the Project Settings (nut icon).
char auth[] = BLYNK_AUTH_TOKEN;
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "YOUR_WIFI_SSID";
char pass[] = "YOUR_WIFI_PASSWORD";
#define DHTPIN D2
// D3
// Uncomment whatever type you're using!
#define DHTTYPE DHT11     // DHT 11
//#define DHTTYPE DHT22   // DHT 22, AM2302, AM2321
//#define DHTTYPE DHT21   // DHT 21, AM2301
DHT dht(DHTPIN, DHTTYPE);
BlynkTimer timer;
// This function sends Arduino's up time every second to Virtual Pin (5).
// In the app, Widget's reading frequency should be set to PUSH. This means
// that you define how often to send data to Blynk App.
WidgetLCD lcd(V1);
BLYNK_WRITE(V0)
{
int value = param.asInt();
Serial.println(value);
if(value == 1)
{
   digitalWrite(D8, HIGH);
   Serial.println("LED ON");
   lcd.print(0, 0, "LIGHT ON- V=12V");
   lcd.print(0, 1, " I=52mA,P=12W  ");
    delay(2000);
    lcd.clear();
```

```
  }
  if(value == 0)
  {
    digitalWrite(D8, LOW);
    Serial.println("LED OFF");
    lcd.print(0, 0, "LIGHT ON, V=12V  ");
   lcd.print(0, 1, " I=0mA,P=0 W  ");
    delay(2000);
    lcd.clear();
  }
}
void sendSensor()
{
  float h = dht.readHumidity();
  float t = dht.readTemperature(); // or dht.readTemperature(true) for Fahrenheit
  if (isnan(h) || isnan(t))
  {
   Serial.println("Failed to read from DHT sensor!");
   return;
  }
  // You can send any value at any time.
  // Please don't send more that 10 values per second.
  Blynk.virtualWrite(V5, t);
  Blynk.virtualWrite(V6, h);
  Serial.println("TEMPERATURE=");
  Serial.print(t);

  Serial.println("HUMIDITY=");
  Serial.print(h);
  if(t > 34)
   {
  // Blynk.email("shameer50@gmail.com", "Alert", "Temperature over 28C!");
   Blynk.logEvent("temp_alert","Temp above 34 degree");
    digitalWrite(D7, HIGH);
     lcd.print(0, 0, "HIGH TEMP");
   lcd.print(0, 1, "HIGH SPEED-FAN  ");
    delay(2000);
    lcd.clear();
    }
    else
    {
     digitalWrite(D7, LOW);
```

```
      lcd.print(0, 0, "NORAMAL TEMP");
    lcd.print(0, 1, "LOW SPEED-FAN  ");
     delay(2000);
     lcd.clear();
    }
}
//int pirValue; // Place to store read PIR Value
void setup()
{
  // Debug console
  Serial.begin(9600);
  pinMode(D8, OUTPUT);
    pinMode(D7, OUTPUT);
      pinMode(D6, OUTPUT);
        pinMode(D5, OUTPUT);
  Blynk.begin(auth, ssid, pass);
  // You can also specify server:
  //Blynk.begin(auth, ssid, pass, "blynk-cloud.com", 8442);
  //Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8442);
  dht.begin();
  // Setup a function to be called every second
  timer.setInterval(1000L, sendSensor);
}
void loop()
{
  Blynk.run();
  timer.run();
     lcd.print(0, 0, "Telecom Monitor");
    lcd.print(0, 1, "   Using IoT             ");
     delay(2000);
     lcd.clear();
    air();
}
void air(void)
{
  unsigned int sensorvalue=0;
  sensorvalue= analogRead(A0);
  Serial.println("Gas =");
  Serial.print(sensorvalue);
 Blynk.virtualWrite(V4,  sensorvalue);
  if(sensorvalue<=65)
  {
```

```
    Blynk.logEvent("fire_alert","FIRE-BE ALERT");
    digitalWrite(D6,HIGH);
    digitalWrite(D5,HIGH);
    lcd.print(0, 0, "FIRE-BE ALERT ");
   lcd.print(0, 1, " EMERGENCY     ");
    delay(2000);
    lcd.clear();
  }
  else
{
digitalWrite(D6,LOW);
digitalWrite(D5,LOW);
lcd.print(0, 0, "NO FIRE ");
lcd.print(0, 1, "***NORMAL***");
delay(2000);
lcd.clear();
}
}
```

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